

ZS-F23

Geometric Fixing of the Type II Trace Normalization:

The Z_2 -Seam $\ln 2$ as the Crossed-Product Additive Entropy Constant, and the Spin- $\frac{1}{2}$ Boundary Origin of de Sitter Type II₁

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GitHub: <https://github.com/KennyKang-git/zspin>

Verification Summary

Verification: 41/41 PASS | Zero Free Parameters. Sole geometric inputs $\mathbf{A} = 35/437$, $\mathbf{Q} = 11$, $(Z, X, Y) = (2, 3, 6)$, with the LOCKED quantum $\ln 2 = \ln(\dim Z)$. **Anti-numerology: AN-F23.1 EXECUTED, PASS** ($p \approx 0.058\% \ll 1\%$). **Status:** Theorems F23.4 (embedding functor), F23.5 (coarse-grained trace matching), F23.6 (closure criterion) **DERIVED**; Theorem F23.7 (edge-mode frame-duality interface) **DERIVED-interpretation** (resolves the v1.2 finite–infinite edge-mode tension via ZS-M30 frame duality through the $\dim(Z) = 2$ seam, not by direct identification). Central Theorems F23.1, F23.2 **DERIVED-CONDITIONAL** on condition C (O-F19.6 Step 1', a genuine OPEN); Reading F23.3 **HYPOTHESIS**. Falsification gates F-F23.1 through F-F23.9. No new free parameter; $(\mathbf{A}, \mathbf{Q}, \dim Z) = (35/437, 11, 2)$ LOCKED.

§0. Abstract

The von Neumann entropy of a Type II crossed-product algebra is defined only up to a **state-independent additive constant**, which Klinger–Kudler–Flam–Satishchandran and the CLPW program fix by hand through a choice of trace normalization or reference state. We show that the Z-Spin corpus fixes this constant geometrically at zero free parameters. (i) The single-cell Z-Spin logical algebra is the finite-dimensional Type I von Neumann algebra $\mathbf{A_ZS} \cong \mathbf{M}_3(\mathbb{C}) \oplus \mathbb{C} \oplus \mathbf{M}_5(\mathbb{C})$ (PROVEN, ZS-Q11), with abelian center $Z(\mathbf{A_ZS}) \cong \mathbb{C}^3$ carrying the three sector projections. (ii) By the Murray–von Neumann universal-embedding theorem, every finite-dimensional von Neumann algebra embeds trace-preservingly into the hyperfinite II₁ factor R . We prove the **Trace-Preserving Central Embedding Functor** (Theorem F23.4, DERIVED): there is a trace-preserving *-monomorphism $\Phi: \mathbf{A_ZS} \rightarrow R$ sending the central projections P_X, P_Z, P_Y to projections of continuous-dimension trace $(3, 2, 6)/11$, with modular-Hamiltonian difference $\Delta K_\Omega(X \rightarrow Y) = -\ln 2 = -\ln(\dim Z)$. This closes the algebraic half of the previously registered gate O-F19.6 Step 1 unconditionally. (iii) The Finite-Register Trace Determinacy and Z_2 -Seam Additive Constant theorems (F23.1, F23.2) then give the per-cross-sector additive constant $\mathbf{c} = \frac{1}{2} \ln 2 = \frac{1}{2} \text{ bit}$, and the residual of O-F19.6 sharpens to the single condition O-F19.6 Step 1' (matching the gravitational modular flow to the Z-Spin equilibrium flow). This paper (v1.2) analyzes whether Step 1' can be closed. We resolve the apparent discrete-versus-continuous spectral objection (the discrete Z-Spin K_Ω is the sector coarse-graining of the continuous gravitational modular operator), prove Theorem F23.5 (the trace-preserving conditional expectation onto the sector center pushes the trace to weights $(3, 2, 6)/11$ and reproduces $\Delta K = -\ln 2$; DERIVED, via Takesaki's conditional-expectation

theorem), and prove Lemma F23.6 reducing Step 1' to a single sharp, falsifiable condition C: the gravitational equilibrium (de Sitter maximum-entropy) state, coarse-grained to the Z-Spin sectors, carries weights (3, 2, 6)/11. Condition C is the Z-Spin emergence dictionary and cannot be established with current corpus or external tools; **Step 1' therefore remains a genuine OPEN**. The anti-numerology Monte Carlo is executed ($p \approx 0.058\% \ll 1\%$). This paper (v1.3) resolves the finite-infinite edge-mode tension in the Z-Spin manner: rather than identifying the $\dim(Z) = 2$ boundary with the infinite Klinger-Kudler-Flam-Satishchandran edge-mode tower directly, Theorem F23.7 recognizes the tension as the gravitational-edge-mode projection of the ZS-F18 finite/infinite Möbius-interface polarity and resolves it by the frame duality of ZS-M30 Theorem 30.1: the edge-mode tower is the external-frame count (Infinity_A) of the same structure whose internal-frame closure (Infinity_B) is one bit of Z-channel capacity, $\ln 2$, the two connected by a frame transformation through the $\dim(Z) = 2$ seam, with controlling theorem the X-Y Tiling Asymmetry (ZS-M6 §5.5, PROVEN; ZS-M17, DERIVED). This upgrades the tension to DERIVED-interpretation and adds the seventh route to the six-route Möbius-trace pattern. 41/41 verification checks PASS; no new free parameter is introduced.

§0.1 Epistemic Status Legend

Status	Definition
PROVEN	Mathematical theorem with complete proof under stated definitions, or machine-precision numerical verification; independent of Z-Spin interpretation.
DERIVED	Quantitative consequence of PROVEN items plus the Z-Spin action; zero free parameters beyond $A = 35/437$.
DERIVED-CONDITIONAL	DERIVED conditional on a stated assumption or upstream OPEN gate, explicitly tracked.
DERIVED-with-bridge	Derived internally and additionally supported by a precise correspondence with an externally PROVEN / published result.
DERIVED-interpretation	A reading of corpus structure, derived from PROVEN/DERIVED items, registered as interpretation (over-determined where multiple independent routes agree); no new free parameter.
VERIFIED	Numerical / computational confirmation strengthening a DERIVED claim at stated precision (here: executed Monte Carlo).
HYPOTHESIS-strong	Multiple independent structural anchors; explicit promotion path documented.
HYPOTHESIS	Motivated reading or conjecture; derivation chain incomplete.
IMPORTED	Result proved externally and used without re-proof; full citation given.
LOCKED	Core constant fixed upstream; no downstream paper may modify.
OPEN	Recognized gap honestly registered; closure path specified where possible.
NON-CLAIM	Explicit declaration of what is NOT asserted; bounds the framework's scope.

§1. Introduction

§1.1 The additive-constant obstruction

The operator-algebraic reformulation of gravitational entropy converts the Type III₁ algebra of quantum-field observables in a gravitational subregion into a Type II crossed product, on which a trace and hence a von Neumann entropy exist, reducing in the semiclassical limit to the generalized entropy

$S_{\text{gen}} = A_{\text{hor}}/(4 G_N) + S_{\text{out}}$ [19, 23, 26]. Kudler-Flam–Leutheusser–Satishchandran established that the dressed algebra is Type II_∞ for a black hole (no maximum-entropy state) and Type II_1 for de Sitter (a maximum-entropy state exists), and that the von Neumann entropy of semiclassical states equals the generalized entropy **up to a state-independent additive constant** [23]. The companion edge-mode paper of Klinger–Kudler-Flam–Satishchandran [26] extends the construction to the complete symmetry group: graviton backreaction perturbs the horizon shape, and at the same perturbative order as the area mode an infinite set of edge-mode charges (region-preserving diffeomorphisms) enters the physical algebra. In all of these constructions the additive constant is fixed by hand — by a choice of trace normalization or reference-state entropy — and sets the zero-point of gravitational entropy.

§1.2 The Z-Spin assets

Z-Spin Cosmology supplies, at zero free parameters, the structure that fixes this constant. **First**, the single-cell logical algebra is the finite-dimensional von Neumann algebra $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ (PROVEN, ZS-Q11 Theorem Q11.A) [2], with abelian center $Z(A_{\text{ZS}}) \cong \mathbb{C}^3$ spanned by the three sector projections P_X, P_Z, P_Y . **Second**, ZS-F19 [1] computed the modular Hamiltonian of the equilibrium state $p_{\text{eq}} = (3, 2, 6)/11$ and found $\Delta K_\Omega(X \rightarrow Y) = -\ln 2$, registering the absolute-scale identification as the three-step gate O-F19.6. **Third**, $\ln 2$ is the Z_2 -seam quantum recurring across the corpus as the Kraus-index parity of the $\dim(Z) = 2$ boundary [4], the Bekenstein–Hawking $-\ln 2$ correction [5], and the Z-bottleneck channel capacity [3]. This paper (v1.1) closes the algebraic half of O-F19.6 Step 1 by a rigorous embedding theorem, sharpens the residual, and positions the result against the edge-mode program.

§1.3 What this paper establishes (v1.3 advances over v1.2)

Theorem F23.4 (Embedding Functor, DERIVED, v1.1). A trace-preserving $*$ -monomorphism $\Phi: A_{\text{ZS}} \rightarrow R$ sends the central projections to projections of trace $(3, 2, 6)/11$ with $\Delta K_\Omega = -\ln 2$; closes the algebraic half of O-F19.6 Step 1 unconditionally.

Spectral resolution (NEW, v1.2, DERIVED). The apparent contradiction between the continuous gravitational modular spectrum (KLS) and the discrete three-point $K_\Omega = -\ln p_{\text{eq}}$ is removed: K_Ω is the modular Hamiltonian of the sector coarse-graining of the gravitational state, discrete by construction.

Theorem F23.5 (Coarse-Grained Trace Matching, NEW, v1.2, DERIVED). Via Takesaki’s conditional-expectation theorem, the trace-preserving conditional expectation E onto the sector center pushes the trace to weights $(3, 2, 6)/11$ and reproduces $\Delta K(X \rightarrow Y) = -\ln 2$ — the maximum-entropy / trace-level matching, independent of any gravitational identification.

Lemma F23.6 (Closure Criterion, NEW, v1.2, DERIVED). O-F19.6 Step 1’ closes if and only if a single sharp, falsifiable condition C holds: the gravitational equilibrium (de Sitter maximum-entropy) state, coarse-grained to the Z-Spin sectors, carries weights $(3, 2, 6)/11$. Condition C is the Z-Spin emergence dictionary.

Verdict (v1.2, honest). Condition C cannot be established with current corpus or external tools — it IS the foundational emergence hypothesis for the gravitational sector. **O-F19.6 Step 1’ remains a genuine OPEN.** The contribution of v1.2 is the precise characterization of the residual (one falsifiable statement), the removal of the spectral objection, and the honest identification of the finite ($\dim 2$) versus infinite

(KKS edge-mode tower) tension. F23.1, F23.2 remain DERIVED-CONDITIONAL on C; F23.4–F23.6 are unconditional (DERIVED).

Theorem F23.7 (Edge-Mode Frame-Duality Interface, NEW, v1.3, DERIVED-interpretation). The finite–infinite edge-mode tension flagged in v1.2 is not a defect but the gravitational-edge-mode projection of the ZS-F18 finite/infinite Möbius-interface polarity. Following the F18 method — do not connect finite and infinite directly; mediate them through the $\dim(Z) = 2$ seam interface — it is resolved by the frame duality of ZS-M30 Theorem 30.1: the infinite KKS edge-mode tower is the external-frame count (Infinity_A) of the same structure whose internal-frame closure (Infinity_B) is one bit of Z-channel capacity, $\ln 2$; the two are connected by a frame transformation, not a direct identification or continuum approximation. The controlling theorem is the X–Y Tiling Asymmetry (ZS-M6 §5.5, PROVEN) with ZS-M17 (DERIVED). This upgrades the v1.2 OPEN tension (NC-F23.9) to DERIVED-interpretation and adds the seventh route to the six-route Möbius-trace pattern (§6.1). The residual sharpens to Condition C-edge (NC-F23.10), a patterned sub-question of Condition C.

§2. Locked Inputs

All quantities are inherited from prior papers. ZS-F23 introduces zero new parameters.

Table 1. Locked inputs for ZS-F23 v1.3.

Quantity	Value	Source	Status
A (geometric impedance)	$35/437 = 0.080092\dots$	ZS-F2	LOCKED
$(Z, X, Y); Q$	$(2, 3, 6); Q = 11$	ZS-F5	PROVEN
$\dim(Z) = j$ multiplicity	$2 = j = 1/2$ (unique)	ZS-M3 Thm 5.1	PROVEN
A_{ZS} (single-cell algebra)	$M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$	ZS-Q11 Q11.A	PROVEN
$Z(A_{\text{ZS}})$ (center)	$\mathbb{C}^3 = \text{span}\{P_X, P_Z, P_Y\}$	ZS-Q11 (this work §4.1)	PROVEN
p_{eq} (equilibrium / modular weights)	$(3, 2, 6)/11$	ZS-Q7, ZS-F19	DERIVED
ΔK_{Ω} ($X \rightarrow Y$ modular diff.)	$-\ln 2 = -\ln(\dim Z)$	ZS-F19	DERIVED
$\psi_{\text{KMS}}(X \rightarrow Y)$	$\frac{1}{2} \ln 2$	ZS-F19	DERIVED
$\ln 2$ (Z_2 -seam / Kraus parity)	$= \ln(\dim Z)$	ZS-A7, ZS-M3, ZS-Q7	DERIVED

Note (code vs equilibrium weighting). The OAQEC code subspace of ZS-Q11 carries block dimensions $(3, 1, 5)$ (the J-odd slots 1 and 9 are stabilized out), used for the code-distance analogue. The modular / additive-constant content of this paper uses the FULL sector weighting $(3, 2, 6)/11$, which is the equilibrium state on which ZS-F19 computed K_{Ω} . The two normalizations are distinct objects and are reconciled in §8 (check F6); the additive constant uses the equilibrium weighting.

§3. The External PROVEN Substrate

§3.1 Tomita–Takesaki theory and the crossed-product structure theorem

Theorem 3.1 (Takesaki 1973, PROVEN, IMPORTED) [14]. Every Type III von Neumann algebra M is isomorphic to the crossed product $N \rtimes_{\sigma} \mathbb{R}$ of a semifinite (Type II $_{\infty}$) algebra N by the modular

automorphism group σ , and N carries a faithful normal semifinite trace τ ; a second crossed product returns Type III (Takesaki duality). The modular flow $\Delta^{\{s\}}$ leaves the algebra invariant (Tomita) [15].

§3.2 The origin of the additive constant

On a Type II factor the trace is unique only up to a positive scalar λ (Murray–von Neumann) [13]. For a state ϕ with density ρ_ϕ ($\phi(\cdot) = \tau(\rho_\phi \cdot)$), rescaling $\tau \rightarrow \lambda\tau$ sends $\rho_\phi \rightarrow \rho_\phi/\lambda$ and

$$S_\tau(\phi) \rightarrow S_\tau(\phi) + \ln \lambda. \quad (1)$$

The entropy is defined up to the additive constant $\ln \lambda$, fixed only by the trace normalization — the precise algebraic origin of the state-independent constant of [23, 26].

§3.3 The current fix-by-hand convention

In the gravitational setting the freedom is fixed by convention: one declares a reference state (empty de Sitter, or the maximally mixed state) to take a chosen entropy [23]. The constant is not supplied by the algebra.

§3.4 The Murray–von Neumann universal-embedding theorem (NEW import)

Theorem 3.4 (Murray–von Neumann 1943; Connes 1976, PROVEN, IMPORTED) [13, 16]. The hyperfinite II_1 factor R is the unique approximately-finite-dimensional II_1 factor and is the smallest infinite-dimensional factor: every finite-dimensional von Neumann algebra (and more generally every AFD algebra) admits a trace-preserving normal $*$ -monomorphism into R . R has projections of every trace in $[0, 1]$ (continuous dimension), and its trace is unique. The infinite tensor product of finite Type I_n factors with the tracial state is R ; with a non-tracial product state it is a Type III factor (Powers; Araki–Woods) [17, 18].

Theorem 3.4 is the new external input enabling the v1.1 embedding theorem (§4.4).

§4. The Finite Type I Collapse and the Embedding Functor

§4.1 A_{ZS} as a finite Type I algebra and its center

ZS-Q11 (Theorem Q11.A, PROVEN) [2] established $A_{\text{ZS}} \cong M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_5(\mathbb{C})$ on the code subspace. A finite direct sum of full matrix algebras is finite Type I; its trace is canonical (the matrix trace, minimal projections trace 1), with no continuous scalar freedom. Its **center** is the abelian algebra

$$Z(A_{\text{ZS}}) = \mathbb{C} \cdot P_X \oplus \mathbb{C} \cdot P_Z \oplus \mathbb{C} \cdot P_Y \cong \mathbb{C}^3, \quad (2)$$

spanned by the three central (sector) projections. The Z-block parity — the Kraus-index parity of the $\dim(Z) = 2$ boundary — carries entropy $\ln 2 = \ln(\dim Z)$ (ZS-Q11 Theorem Q11.B, ZS-A7) [2, 4].

§4.2 Theorem F23.1 (Finite-Register Trace Determinacy)

Statement. If the emergent gravitational Type II factor M_{grav} is the AFD limit of the multi-cell register $\otimes_{\{v\}} A_{\text{ZS},v}$, the trace-normalization scalar λ — the sole source of the additive-constant freedom (§3.2) — is fixed by the finite canonical trace of A_{ZS} and is not free. **Proof:** §4.4 (F23.4) + Theorem 3.4.

Status: **DERIVED-CONDITIONAL** on O-F19.6 Step 1' (§4.5); the embedding core is **DERIVED** (F23.4).

§4.3 Theorem F23.2 (Z_2 -Seam Additive Constant)

Statement. Under F23.1, the state-independent additive constant per cross-sector ($X \rightarrow Y$) modular transition is

$$c = \frac{1}{2} |\Delta K_\Omega| = \frac{1}{2} \ln 2 = \frac{1}{2} \text{ bit}, \quad (3)$$

with $\Delta K_\Omega = \ln(p_X/p_Y) = \ln(3/6) = -\ln 2$ the modular-Hamiltonian difference of $p_{eq} = (3, 2, 6)/11$ (ZS-F19) [1], equal to the boundary parity $-\ln(\dim Z)$ (ZS-A7) [4] and the Z-bottleneck per-transition entropy (ZS-Q7) [3]. **Status:** magnitude $\frac{1}{2} \ln 2$ **DERIVED**; identification with the gravitational additive constant **DERIVED-CONDITIONAL** on O-F19.6 Step 1'.

§4.4 Theorem F23.4 (Trace-Preserving Central Embedding Functor) — NEW

Statement. There is a unital, normal, trace-preserving *-monomorphism $\Phi: A_{ZS} \rightarrow R$ into the hyperfinite II_1 factor such that the central projections map to mutually orthogonal projections $e_X = \Phi(P_X)$, $e_Z = \Phi(P_Z)$, $e_Y = \Phi(P_Y)$ with

$$\tau_R(e_X) = 3/11, \tau_R(e_Z) = 2/11, \tau_R(e_Y) = 6/11, e_X + e_Z + e_Y = 1, \quad (4)$$

and the modular Hamiltonian of the equilibrium state on $Z(A_{ZS})$, $K_\Omega = -\ln p_{eq}$, satisfies $\Delta K_\Omega(X \rightarrow Y) = -\ln 2 = -\ln(\dim Z)$. Φ is functorial: any trace-preserving inclusion $A_{ZS} \rightarrow A_{ZS} \otimes M_k$ of register cells extends to R compatibly with Φ .

Proof. By Theorem 3.4 [13, 16], R contains a trace-preserving normal *-monomorphic copy of every finite-dimensional von Neumann algebra; apply this to A_{ZS} with the trace assigning the central projections the equilibrium weights $p_{eq} = (3, 2, 6)/11$. R has projections of every trace in $[0, 1]$ (continuous dimension), so e_X, e_Z, e_Y of traces $3/11, 2/11, 6/11$ exist and are mutually orthogonal with sum 1 because Φ is a unital *-homomorphism on the abelian projection system $Z(A_{ZS})$. The modular-Hamiltonian values are the ZS-F19 computation [1], invariant under Φ because Φ is trace-preserving. Functoriality is the compatibility of the AF inductive system $\otimes_v A_{ZS,v}$ with the universal embedding.

Significance. F23.4 closes the **algebraic half** of O-F19.6 Step 1 unconditionally: the central / block projections of A_{ZS} map to projections in a Type II_1 factor preserving trace and the modular difference $-\ln 2$. The embedding, the projection traces $(3, 2, 6)/11$, and the value $\frac{1}{2} \ln 2$ are thereby **DERIVED**, not conditional. Status: **DERIVED**.

§4.5 The sharpened residual: O-F19.6 Step 1' (modular-flow matching)

F23.4 does not establish that the abstract target R is the SPECIFIC gravitational subregion algebra M_{grav} of [19, 23, 26], i.e. that e_X, e_Z, e_Y are the physical projections of M_{grav} and that the Z -Spin equilibrium flow $\sigma_t^\wedge \Omega = \exp(-itK_\Omega)$ equals the gravitational modular flow σ_t^{grav} . The residual of O-F19.6 therefore sharpens to the single condition:

O-F19.6 Step 1' (modular-flow matching): $\sigma_t^{\wedge \text{grav}}$ restricted to the three-projection subalgebra equals $\sigma_t^{\wedge \Omega}$ (equivalently, the gravitational modular Hamiltonian restricted to the sector center equals $-\ln p_{\text{eq}}$).

Two HYPOTHESIS-strong promotion paths are registered. **(a) Observer-clock identification:** in De Vuyst–Höhn [22, 23] the observer carries a clock whose Hamiltonian generates the crossed-product flow; identifying the dim-2 Z-sector (the Z-bottleneck) with this observer clock makes $\sigma_t^{\wedge \Omega}$ the modular flow by construction. **(b) Edge-mode identification:** in Klinger–Kudler-Flam–Satishchandran [26] the complete physical algebra includes horizon shape (edge-mode) charges; identifying the central projection P_Z with the edge-mode/boundary sector makes e_Z the gravitational edge-mode projection. Either path closes Step 1' and lifts F23.1, F23.2 to full DERIVED. **Status of Step 1': OPEN.** The two physical paths (observer-clock; edge-mode) are analyzed in §4.6–§4.8, where Step 1' is reduced to a single sharp condition.

§4.6 Theorem F23.5 (Coarse-Grained Trace Matching) — NEW (v1.2)

Before the gravitational identification, the trace-level content of Step 1' is established unconditionally. Let $\mathcal{E}: \mathbf{R} \rightarrow \mathcal{E}(\mathbf{R}) = \text{span}\{e_X, e_Z, e_Y\}$ be the conditional expectation onto the abelian sector subalgebra generated by the embedded central projections (F23.4).

Statement. A normal, trace-preserving conditional expectation $\mathcal{E}$ onto the sector center exists, and

$$\mathcal{E}^*(\tau_{\mathbf{R}}) = (\tau_{\mathbf{R}}(e_X), \tau_{\mathbf{R}}(e_Z), \tau_{\mathbf{R}}(e_Y)) = (3, 2, 6)/11 = p_{\text{eq}}, \quad (5)$$

with coarse-grained modular Hamiltonian $K = -\ln \mathcal{E}^*(\tau_{\mathbf{R}})$ giving $\Delta K(X \rightarrow Y) = \ln(3/6) = -\ln 2 = -\ln(\dim Z)$. **Proof.** Because $\mathbf{R}$ carries a finite trace, its modular automorphism group on the tracial state is trivial; by Takesaki's theorem [29] a normal conditional expectation onto a von Neumann subalgebra exists iff the subalgebra is invariant under the modular flow, which is automatic here, and the expectation is trace-preserving. The pushforward weights are the F23.4 projection traces (Eq. 4). **Status: DERIVED.** This is the maximum-entropy / trace-level matching: the additive constant $\frac{1}{2} \ln 2$ is the coarse-grained-trace value in $\mathbf{R}$, independent of any gravitational identification.

§4.7 Lemma F23.6 (Closure Criterion for Step 1') and the spectral resolution — NEW (v1.2)

Spectral resolution (DERIVED clarification). KLS [23] found that the gravitational modular Hamiltonian has continuous spectrum $\mathbb{R}$ (Type II_∞) or bounded spectrum (Type II_1 , de Sitter), whereas $K_\Omega = -\ln p_{\text{eq}}$ is a discrete three-point operator. There is no contradiction: K_Ω is the modular Hamiltonian of the coarse-grained three-sector state $\mathcal{E}^*(\omega)$, which is discrete by construction (three outcomes), regardless of the continuous spectrum of the full modular operator. The discrete Z-Spin modular structure is the sector coarse-graining of the continuous gravitational one.

Statement (closure criterion). Let ω_{grav} be the gravitational equilibrium (KMS / maximum-entropy) state and $\mathcal{E}$ the sector coarse-graining. On the center the literal modular flows both fix the projections (the center is modular-invariant), so Step 1' reduces to matching the coarse-grained sector weights. Hence O-F19.6 Step 1' closes — and F23.1, F23.2 become DERIVED — if and only if

$$\text{Condition C: } \mathcal{E}^*(\omega_{\text{grav}}) = p_{\text{eq}} = (3, 2, 6)/11. \quad (6)$$

Because the de Sitter maximum-entropy state is the tracial state of the Type II₁ factor, Condition C is equivalent to the statement that the empty-de Sitter maximum-entropy state, coarse-grained to the Z-Spin sectors ($X \leftrightarrow \Lambda$ CDM macro, $Z \leftrightarrow$ boundary, $Y \leftrightarrow$ SM micro), carries weights (3, 2, 6)/11. **Status:** the reduction is **DERIVED**; Condition C is the Z-Spin emergence dictionary and is **OPEN**.

§4.8 Verdict on Step 1': a genuine OPEN, sharply characterized

The deep-exploration analysis converges to a single residual node: Condition C, which is the Z-Spin emergence dictionary for the gravitational sector. This cannot be established with the current corpus or any imported external theorem — it is the foundational emergence hypothesis itself. **O-F19.6 Step 1' therefore remains a genuine OPEN** (a confirmation that the present tools cannot close it). The value of the analysis is threefold: (i) the discrete-versus-continuous spectral objection is removed (§4.7); (ii) the coarse-grained trace matching is proved (F23.5); and (iii) the gate is reduced to a single sharp, falsifiable statement (F23.6): a future microscopic computation or measurement of the gravitational sector weights either equals (3, 2, 6)/11, closing the gate and lifting F23.1–F23.2 to DERIVED, or does not, falsifying ZS-F23 (gate F-F23.8). Per the corpus epistemology, a convergent exploration whose core node remains OPEN is itself a registered asset: the precise location of the boundary between what is DERIVED (F23.4–F23.6) and what is hypothetical (Condition C). **Status: OPEN** (sharply characterized).

§5. The Spin-½ Boundary and de Sitter Type II₁

§5.1 The $\dim(Z) = 2 = j = \frac{1}{2}$ spinor boundary

ZS-M3 (Theorem 5.1, PROVEN) [5] established that $j = 1/2$ is the unique half-integer giving $(Z, X, Y) = (2, 3, 6)$; the boundary is a spinor with $\dim(Z) = 2 = j = 1/2$. ZS-A7 (§3) [4] showed the boundary holonomy B_Z closes at 4π in the $j = 1/2$ representation, and the $-\ln 2$ of the Bekenstein–Hawking correction is the Kraus-index parity entropy. The register symmetry is $D_4 = \langle J, J_Z \rangle$ (PROVEN) [2].

§5.2 The 2026 spin-energy result for the de Sitter static patch

The de Sitter static-patch analysis [27] finds that a proper account of the observer's rotational ("spin") energy and non-gravitational binding energy forces the algebra to be Type II₁ with a maximum-entropy state (empty de Sitter); the covariant-observer construction realizes the observer as a superposition of geodesics carrying an $L^2(\text{SO}(1, d))$ quantum reference frame.

§5.3 Reading F23.3

Reading. The type-determining boundary degree of freedom is the $\dim(Z) = 2$ spinor; we read the external requirement that the observer's spin energy forces Type II₁ as the statement that this spinor is responsible for the conversion, with the boundedness of Type II₁ reflecting the finite $\mathbf{Q} = 11$ register. The cyclic-cosmology entropy $S_{\{Z_2\}} = 6\pi/A$ (ZS-U8) [10] is the Z-Spin counterpart of the de Sitter maximum-entropy state. **Honest boundary (NC-F23.3):** the external "spin energy" is rotational kinetic energy, not a priori a $j = 1/2$ representation; this is a structural resonance, not a derivation. Status: **HYPOTHESIS**.

§5.4 The observer-disagreement bound and $\frac{1}{2} \ln 2$

A complementary 2026 result bounds how much quantum observers can disagree about subsystem entropy (arXiv:2603.23598). The Z-Spin per-cross-sector value $c = \frac{1}{2} \ln 2 = \frac{1}{2} \text{ bit}$ (F23.2) is the candidate Z-Spin value for one quantum of this disagreement. Status: **HYPOTHESIS-strong**.

§6. Position Relative to the Edge-Mode Program

The following table places ZS-F23 slot-by-slot against the external Type II crossed-product program of Klinger–Kudler-Flam–Satishchandran (KKS) [26] and Kudler-Flam–Leutheusser–Satishchandran (KLS) [23] / CLPW [19]. The purpose is to make explicit which element of the external construction each Z-Spin object touches. The central result of §4 is the row “Additive constant”: F23’s $\frac{1}{2} \ln 2$ occupies exactly the normalization slot the external program leaves to convention.

Table 2. Slot-by-slot comparison with the external Type II crossed-product / edge-mode program.

Element	External (KKS [26] / KLS [23] / CLPW [19])	ZS-F23 (this work)	Same slot?
Algebra type	Type II $_{\infty}$ (BH) / Type II $_1$ (de Sitter), from the modular crossed product of a Type III subregion algebra.	AFD limit of $\otimes_v A_{\text{ZS}}$ (A_{ZS} finite Type I); II $_1$ (tracial state) / II $_{\infty}$ or III (non-tracial), via the same modular crossed product.	YES
Entropy formula	$S_{\text{vN}} = S_{\text{gen}} + \text{const}$ (generalized entropy up to a state-independent constant) [23].	$S_{\text{vN}} = S_{\text{gen}} + c$, identical structure.	YES
The additive constant	State-independent; FIXED BY HAND via trace normalization or reference-state entropy (free).	$c = \frac{1}{2} \ln 2 = \frac{1}{2} \ln(\dim Z)$ per cross-sector, FIXED by the finite-register canonical trace (F23.1–F23.4).	YES (the claim)
Origin of the constant	The Type II trace scalar λ (entropy shift $\ln \lambda$); not fixed by the algebra.	Finite Type I canonical trace (Murray–von Neumann); the scalar λ is pinned, no continuous freedom (§3.2, F23.4).	YES
Edge / shape modes	Infinite set of horizon shape charges (region-preserving diffeomorphisms) = the complete symmetry group [26].	The Z-sector boundary ($\dim 2 = j = 1/2$ spinor); central projection $P_Z \rightarrow e_Z$.	HYPOTHESIS
Edge-mode dimension	Infinite-dimensional tower (one charge per region-preserving diffeomorphism) [26].	External-frame count (Infinity $_A$) of a structure whose internal-frame closure (Infinity $_B$) is the $\dim(Z) = 2 \text{ bit}$ ($\ln 2$); connected by frame transformation through the $\dim(Z) = 2$ seam.	DERIVED-interp (frame duality, F23.7)
Observer / clock	Observer energy bounded below $\rightarrow$ II $_1$; clock Hamiltonian generates the crossed-product flow [22, 23].	Z-bottleneck / Z-sector as observer clock; $\sigma_t^* \Omega$ the equilibrium modular flow (O-F19.6 Step 1’).	HYPOTHESIS-strong
Maximum-entropy state	Empty de Sitter (Type II $_1$ max-entropy state) [23].	Cyclic-cosmology entropy $S_{\{Z_2\}} = 6\pi/A$ (ZS-U8) [10].	HYPOTHESIS

The table answers the external reader’s question directly: ZS-F23 touches the **additive-constant / trace-normalization slot** of the generalized-entropy formula, the slot the external program fixes by convention. The edge-mode and observer-clock rows are the physical content of Condition C (§4.7); the row “Edge-mode dimension” is resolved in §6.1 by frame duality (Theorem F23.7), upgrading the v1.2 OPEN tension to DERIVED-interpretation.

§6.1 Theorem F23.7 (Edge-Mode Frame-Duality Interface) — the Z-Spin resolution of the finite–infinite tension

The principal tension between ZS-F23 and the edge-mode program is dimensional: the Klinger–Kudler-Flam–Satishchandran physical algebra carries an **infinite** tower of horizon shape (edge-mode) charges, one per region-preserving diffeomorphism [26], whereas the Z-sector boundary is **finite**, $\dim(Z) = 2$. A naive identification — declaring the Z-sector the per-link tensor factor of the infinite tower — is a category error: it forces a finite object to equal an infinite one by approximation. The Z-Spin corpus prescribes a different move.

The F18 method. ZS-F18 establishes that across all twelve Millennium/Crisis problems the recurring structure is the finite/infinite Möbius-interface polarity: X-finite/local/decidable information communicates with Y-infinite/non-local/undecidable information **only through the $\dim(Z) = 2$ seam interface**, never directly [F18 §5]. The edge-mode tension is the gravitational-edge-mode projection of exactly this polarity. One does not connect finite and infinite; one mediates them through the seam.

Statement (Theorem F23.7, DERIVED-interpretation). Under the frame duality of ZS-M30 Theorem 30.1, the infinite KKS edge-mode tower is the external-frame count (Infinity_A) of the same structure whose internal-frame closure (Infinity_B) is one bit of Z-channel capacity, $\ln 2$, equal to one full Möbius pass $2c = 2 \cdot (\frac{1}{2} \ln 2) = \ln 2$. The two are connected by a frame transformation through the $\dim(Z) = 2$ seam — the mediating interface — not by a direct identification or a continuum approximation. The controlling theorem is the X–Y Tiling Asymmetry (ZS-M6 §5.5, PROVEN): the truncated octahedron tiles $\mathbb{R}^3$ uniquely (the unbounded tiling = the external count of boundary modes; X-side continuum-emergent), while the truncated icosahedron cannot tile (I_h forbids it; Y-side structurally discrete), and the Z-sector mediates with capacity $\ln 2$ — with operational consequence the Tiling Continuum Convergence Theorem (ZS-M17 Thm M17.1, DERIVED). Discrete/continuous — equivalently finite-tower/continuous-boundary — are therefore co-resident sectoral attributes of the (X, Y) decomposition (ZS-F18 §7.4, sixth polarity row), not opposed primitives.

Basis and status. F23.7 is the seventh route of the six-route Möbius-trace pattern of ZS-M30 §3 (the gravitational edge-mode route, joining Riemann zeros $\leftrightarrow z^*$, cosmological cycle index $\leftrightarrow A$, the BV–BFV ghost tower $\leftrightarrow 4\pi$ closure, FFPP compression $\leftrightarrow W_0(-i\pi/2)$, Z-Spin mediation outcomes $\leftrightarrow \ln 2$ per pass, and the Wilson-cycle ergodic $1/2$ limit). It inherits the DERIVED-interpretation-strong standing of ZS-M30 Theorem 30.1 and ZS-F18 §7.4, anchored by the PROVEN ZS-M6 §5.5 and the DERIVED ZS-M17. Status: **DERIVED-interpretation**. This upgrades the v1.2 OPEN tension (NC-F23.9): the finite-versus-infinite mismatch is not a contradiction but frame duality.

What F23.7 does and does not do (honest boundary). F23.7 resolves the **conceptual** finite–infinite tension: a finite $\dim=2$ boundary and an infinite edge-mode tower are the internal- and external-frame faces of one structure, mediated by the seam. It does NOT by itself close Condition C. The residual sharpens to **Condition C-edge** (NC-F23.10): whether the specific gravitational/KKS edge-mode external count instantiates the X–Y Tiling Asymmetry frame transformation with the $\ln 2$ invariant. This is now a patterned sub-question of Condition C — an instance of an over-determined corpus pattern rather than a bare gap — which modestly reduces the scope of the Step 1' OPEN.

§7. Anti-Numerology and Zero-Free-Parameter Audit

§7.1 Zero-free-parameter audit

Every quantity traces to a LOCKED, PROVEN, DERIVED, or IMPORTED item: $A = 35/437$ (ZS-F2), $Q = 11$ and $(Z, X, Y) = (2, 3, 6)$ (ZS-F5), A_ZS and its center (ZS-Q11), p_eq and the master-equation eigenvalues $\lambda(\lambda + 2A/Q)(\lambda + A) = 0$ (ZS-Q7), $\Delta K_Q = -\ln 2$ (ZS-F19), $\ln 2 = \ln(\dim Z)$ (ZS-A7, ZS-M3), and the embedding target R (Murray–von Neumann). No fudge factor is introduced. Negative control: the additive constant is independent of A (it tracks $\dim(Z)$).

§7.2 AN-F23.1: pre-registered joint structural-signature Monte Carlo (EXECUTED)

Within ZS-F23 the value $c = \frac{1}{2} \ln 2$ is **deterministic** given the LOCKED sector geometry $(2, 3, 6)$ and the LOCKED sector roles (X, Y matter; Z the bottleneck boundary): there is no fitting room, and no measured additive constant exists to fit to (the value is a prediction, not a fit). The Monte Carlo therefore tests the robustness of the value-producing structure, not a data fit.

Discriminator. The additive constant equals $\frac{1}{2} \ln(\dim Z)$ because the locked geometry satisfies the joint structural signature: $S1$ (sum = $Q = 11$, ZS-F5), $S2$ (product = $36 = 6^2$, ZS-F5), $S3$ ($d_Y = d_X \cdot d_Z$, so $|\ln(d_X/d_Y)| = \ln(\dim Z)$), and $S4$ ($d_Z = 2 = j = 1/2$, ZS-M3). Among random ordered integer triples in $\{1, \dots, 12\}^3$ (1728 triples) the executed counts are:

Pre-registered signature	Hits	p	Surviving triples (Z, X, Y)
$S1$ (sum = 11)	45	2.604%	—
$S1 \wedge S2$ (sum 11, product 36)	6	0.347%	all 6 permutations of $(2, 3, 6)$
$S1 \wedge S2 \wedge S3$ ($+ d_Y = d_X \cdot d_Z$)	2	0.116%	$(2, 3, 6), (3, 2, 6)$
$S1 \wedge S2 \wedge S3 \wedge S4$ ($+ d_Z = 2$)	1	0.058%	$(2, 3, 6)$ — unique

Result: $p \approx 0.058\% \ll 1\%$ — PASS. The full joint signature selects $(2, 3, 6)$ uniquely. **Inheritance caveat (honest).** Most of the specialness (down to $p \approx 0.347\%$) is the ZS-F5 register geometry (sum and product), which carries its own upstream anti-numerology; ZS-F23's own marginal contribution is the additive-constant identity $S3$, which given the geometry holds for 2 of the 6 orderings and is selected by the LOCKED matter/bottleneck roles. Within the locked triple two coincidences $|\Delta K| = \ln(d_i)$ exist ($X \rightarrow Y / d_Z$ and $Z \rightarrow Y / d_X$); the physical role assignment selects the matter-matter / bottleneck-parity one. The v1.0 single-identity test over a geometry-agnostic ensemble was marginal ($p \approx 9\%$); the corrected pre-registration uses the full joint signature. Per the corpus scoring rule, AN-F23.1 is now EXECUTED with $p \ll 1\%$, removing the v1.0 pending status; F23.1–F23.2 remain DERIVED-CONDITIONAL on Step 1' and are registered at HYPOTHESIS-strong overall.

§8. Cross-Paper Consistency (Version-Conflict Check)

ZS-F23 is strictly additive: no new constant, none modified. The chain A (ZS-F2) $\rightarrow Q$, (Z, X, Y) (ZS-F5) $\rightarrow \ln 2$, $\dim(Z) = 2$ (ZS-M3, ZS-A7) $\rightarrow p_eq$, λ -factorization (ZS-Q7) $\rightarrow \Delta K_Q = -\ln 2$ (ZS-F19) $\rightarrow A_ZS$ and $Z(A_ZS)$ (ZS-Q11) is internally consistent (Appendix A, Category F).

Code/equilibrium reconciliation (F6): the OAQEC code weighting $(3, 1, 5)/9$ (slots 1, 9 stabilized out) and the modular weighting $(3, 2, 6)/11$ are distinct; the additive constant uses the equilibrium $(3, 2, 6)/11$, consistent with ZS-F19. The i-tetration fixed point z^* and $|f'(z^*)| = 0.892$ (ZS-M1, ZS-F3) and their

downstream uses (ZS-S1, ZS-U1) are orthogonal to the modular-trace content and unaffected. The cross-paper audit across 12 upstream references PASSES.

§9. Consistency with Observation

The additive constant $c = \frac{1}{2} \ln 2 = 0.5$ bit is below any current sensitivity for gravitational entropy (not directly measured); ZS-F23 makes no new cosmological prediction and cannot conflict with the Planck 2018 Λ CDM parameters [28]. The de Sitter Type II₁ maximum-entropy state, read as $S_{\{Z_2\}} = 6\pi/A$ (ZS-U8) [10], is consistent with the X-sector Λ CDM fit (ZS-U4, ZS-U6) without modification. No observational tension is introduced.

§10. Falsification Gates

Table 3. Multilayered falsification gates for ZS-F23 v1.3.

Gate	Layer	Falsification condition	Status
F-F23.1	Math / theoretical (immediate rejection)	$A_ZS \cong M_3 \oplus \mathbb{C} \oplus M_5$ (ZS-Q11) shown not to embed trace-preservingly into R , or Takesaki / Murray–von Neumann structure misapplied.	PASSING
F-F23.2	Theoretical (demotion)	O-F19.6 Step 1' (modular-flow matching) shown impossible $\rightarrow$ F23.1, F23.2 demoted to NON-CLAIM (F23.4 unaffected, being unconditional).	OPEN
F-F23.3	Simulation / consistency (revision)	Joint-signature MC fails to select (2,3,6), or a value other than $\ln(\dim Z)$ is forced. EXECUTED: $p \approx 0.058\%$.	PASSING
F-F23.4	Cross-paper (revision)	$\Delta K_Q \neq -\ln 2$, A_ZS structure changes, or $Z(A_ZS) \neq \mathbb{C}^3$ under any corpus-consistent recomputation.	PASSING
F-F23.5	Observational / external (rejection)	Additive constant independently fixed (microscopic computation) to a value incompatible with $\frac{1}{2} \ln(\dim Z)$.	OPEN (external)
F-F23.6	External (retraction of reading)	de Sitter Type II ₁ “spin energy” or the edge-mode sector shown unrelated to any spinor / boundary structure $\rightarrow$ Reading F23.3 and the edge-mode row retracted (F23.1, F23.2, F23.4 unaffected).	OPEN (external)
F-F23.7	Structural (blocking)	$\dim(Z) \neq 2$ in the physical sector decomposition $\rightarrow$ entire bridge collapses (shared with F-Q7.6).	BLOCKING
F-F23.8	Closure criterion (decisive, external)	Condition C fails: a microscopic computation or measurement of the gravitational equilibrium (de Sitter max-entropy) state, coarse-grained to the Z-Spin sectors, gives weights $\neq (3, 2, 6)/11 \rightarrow$ Step 1' cannot close and F23.1, F23.2 are demoted to NON-CLAIM (F23.4–F23.6 unaffected).	OPEN (decisive)
F-F23.9	Frame-duality interface (revision)	The X–Y Tiling Asymmetry (ZS-M6 §5.5) or ZS-M30 Theorem 30.1 is retracted, or the gravitational edge-mode count is shown NOT to be the external-frame manifestation of a $\ln 2$ internal closure $\rightarrow$ Theorem F23.7 retracted, NC-F23.9 reverts to OPEN tension (F23.1–F23.6 unaffected).	PASSING

§11. Non-Claims

NC-F23.1. ZS-F23 does not derive the generalized-entropy formula itself; it addresses only the additive constant.

NC-F23.2. F23.1 and F23.2 are DERIVED-CONDITIONAL on Condition C (O-F19.6 Step 1', a genuine OPEN); the embedding functor F23.4, the coarse-grained trace matching F23.5, and the closure criterion F23.6 are unconditional (DERIVED).

NC-F23.3. Reading F23.3 (spin-energy $\leftrightarrow$ spinor boundary) and the edge-mode identification (Table 2) are HYPOTHESIS; no quantity is derived from them.

NC-F23.4. No new free parameter; $(A, Q, \dim Z) = (35/437, 11, 2)$ LOCKED. The embedding target R introduces no parameter (it is a uniqueness theorem).

NC-F23.5. ZS-F23 does not assert that the abstract embedding target R IS the specific gravitational subregion algebra; that identification is Condition C (OPEN).

NC-F23.6. No claim about black-hole microstate counting; the contribution is the trace normalization, not the microstates (cf. Witten [20]).

NC-F23.7. The AN-F23.1 specialness is, by the inheritance caveat (§7.2), largely upstream (ZS-F5/M3) geometry; ZS-F23's own contribution is the deterministic, A-independent value of c.

NC-F23.8. ZS-F23 does NOT claim that Step 1' is closed. v1.2 proves it reduces to Condition C and that Condition C is the Z-Spin emergence dictionary; Step 1' is a genuine OPEN. The paper's contribution here is the characterization of the residual, not its closure.

NC-F23.9. The finite-versus-infinite edge-mode mismatch is NOT claimed to be a contradiction or a defect; v1.3 (Theorem F23.7) resolves it as frame duality (the infinite tower and the dim-2 seam are external/internal-frame faces of one structure). What is NOT claimed is a direct identification $\dim(Z) = 2 = (\text{infinite tower})$ by approximation — that move (the v1.2 $\otimes$ links reading) is explicitly rejected as a category error.

NC-F23.10. Theorem F23.7 resolves the conceptual finite–infinite tension but does NOT close Condition C-edge: whether the specific gravitational/KKS edge-mode external count instantiates the X–Y Tiling Asymmetry frame transformation with the $\ln 2$ invariant. Condition C-edge is a patterned sub-question of Condition C (an instance of the over-determined Möbius-trace pattern), and remains OPEN.

§12. Conclusion

The additive constant of Type II gravitational entropy is, in the external program, a free conventional input; the Z-Spin finite-register structure fixes its value at $\frac{1}{2} \ln 2 = \frac{1}{2} \ln(\dim Z)$ with zero free parameters. v1.1 established the trace-preserving central embedding functor (F23.4, DERIVED) and reduced the absolute-scale gate to the single modular-flow-matching condition O-F19.6 Step 1'. This paper (v1.2) asked whether Step 1' can be closed. The answer is honest and precise. We removed the apparent discrete-versus-continuous spectral objection by identifying K_Ω as the sector coarse-graining of the continuous gravitational modular operator; we proved the coarse-grained trace matching (F23.5, DERIVED, via Takesaki's conditional-expectation theorem), establishing that the value $\frac{1}{2} \ln 2$ is the

maximum-entropy / trace-level matching in the embedding target; and we reduced Step 1' to a single sharp, falsifiable statement — Condition C: the de Sitter maximum-entropy state, coarse-grained to the Z-Spin sectors, carries weights (3, 2, 6)/11 (Lemma F23.6, DERIVED). Condition C is the Z-Spin emergence dictionary and cannot be established with current corpus or imported external tools; **O-F19.6 Step 1' therefore remains a genuine OPEN**. This negative result is itself a registered corpus asset: the exploration converges, the spectral objection is gone, the trace-level matching is proved, and the residual is now one falsifiable condition (gate F-F23.8) rather than an open-ended program. F23.1, F23.2 remain DERIVED-CONDITIONAL on Condition C; F23.4–F23.6 stand unconditionally. v1.3 takes the sharpest remaining tension — the finite dim-2 Z-sector versus the infinite KKS edge-mode tower — and resolves it in the characteristic Z-Spin manner (Theorem F23.7): not by forcing finite and infinite together, but by mediating them through the $\dim(Z) = 2$ seam interface, where the infinite tower is the external-frame count and the dim-2 closure is the internal-frame bit ($\ln 2$), connected by the frame transformation of ZS-M30 Theorem 30.1 and controlled by the X–Y Tiling Asymmetry of ZS-M6. The mismatch is thereby revealed as frame duality, not contradiction — a seventh instance of an over-determined corpus pattern — and the residual reduces to the patterned sub-question Condition C-edge. Z-Spin's contribution, as elsewhere, is the removal of a free parameter; here, additionally, the dissolution of an apparent contradiction into a frame-dual interface, with the boundary of the unknown drawn exactly.

Acknowledgements & Code Availability

This work was developed within the Z-Spin Cosmology Collaboration through free- and deep-exploration sessions in June 2026. The author thanks AI tools (Anthropic Claude, OpenAI ChatGPT, Google Gemini) for verification, cross-paper consistency auditing, and exploratory analysis, and retains full responsibility for all scientific content. The verification suite `zs_f23_verify_v1_3.py` (41 checks, categories A–H, 50-digit mpmath precision; executed joint-signature Monte-Carlo gate AN-F23.1) will be publicly available at <https://github.com/KennyKang-git/zspin>.

Appendix A. Verification Suite (41 checks)

Table A1. Verification suite mapping, `zs_f23_verify_v1_3.py`.

Category	Tests	Key check
A. Locked-input cross-checks	5/5	$A = 35/437$; $Q = 11$; $(Z, X, Y) = (2, 3, 6)$; $\ln 2 = \ln(\dim Z)$; $p_{eq} = (3, 2, 6)/11$.
B. Finite Type I / AFD / center / embedding	7/7	$A_{ZS} \cong M_3 \oplus \mathbb{C} \oplus M_5$; $\dim 9 = 1+3+5$; minimal-projection trace 1; finite Type I; AFD trace-preserving inclusion; $Z(A_{ZS}) \cong \mathbb{C}^3$ (3 central projections); embedding sends central projections to traces (3,2,6)/11 summing to 1.
C. Modular and additive constant	5/5	$K_{\Omega} = -\ln p_{eq}$; $\Delta K_{\Omega} = -\ln 2$; $\psi_{KMS} = \frac{1}{2} \ln 2$, $\tanh(2\psi) = 3/5$; $S \rightarrow S + \ln \lambda$; $c = \frac{1}{2} \ln 2$.
D. External substrate consistency	4/4	Takesaki III $= II_{\infty} \rtimes \mathbb{R}$; Murray–von Neumann/Connes trace uniqueness and universal embedding into R ; hyperfinite $II_1 = \text{ITP of type I}_n$ (tracial); CLPW/KLS additive-constant statement.
E. Anti-numerology	5/5	1-bit entropy $= \ln(\dim Z)$; negative control $c \perp A$; $S(I/9) = \ln 9$; AN-F23.1 single-identity (marginal, documented); AN-F23.1 joint signature selects (2,3,6) uniquely, $p \approx 0.058\%$.

Category	Tests	Key check
F. Cross-paper consistency	6/6	ΔK_Ω (ZS-F19); eigenvalues $\{0, -2A/Q, -A\}$, $p_{eq} = (3,2,6)/11$, capacity $\leq \ln 2$ (ZS-Q7); A_{ZS} (ZS-Q11); $\ln 2 =$ Kraus parity (ZS-A7); code $(3,1,5)/9$ vs equilibrium $(3,2,6)/11$ reconciliation; $(A, Q, \dim Z)$ LOCKED, z^* untouched.
G. Coarse-graining / Step 1' closure (v1.2)	5/5	Conditional expectation E onto sector center is unital/idempotent/trace-preserving/bimodule with $E^*(\tau) = (3,2,6)/11$; coarse-grained $\Delta K = -\ln 2$; continuous-spectrum density $\rightarrow$ discrete sector weights (CP, sum 1); Condition C non-vacuous; max-entropy = tracial, $S(I/11) = \ln 11$.
H. Frame-duality interface (v1.3)	4/4	Infinity_B closure quantum $= 2c = \ln(\dim Z) = \ln 2$; X-Y Tiling Asymmetry structural data ($\chi(TO) = \chi(TI) = 2$; $5 \notin$ crystallographic $\{1,2,3,4,6\}$ so TI cannot tile, $3,4 \in$ so TO tiles); capacity $\dim(Z)$ -determined not count-determined (rank ≤ 2 for tower sizes 2..500); seventh Möbius route quantum $= \ln 2$, $c = \text{route}/2$.

Appendix B. Entropy Shift, AFD Lift, and the Embedding Functor

(B.1) Entropy shift. For a trace τ on a Type II factor and a state ϕ with density ρ_ϕ , rescaling $\tau \rightarrow \lambda\tau$ gives density ρ_ϕ/λ and $S_{\{\lambda\tau\}}(\phi) = -\lambda\tau((\rho_\phi/\lambda) \ln(\rho_\phi/\lambda)) = S_\tau(\phi) + \ln \lambda$ (using $\tau(\rho_\phi) = 1$). The additive constant is exactly $\ln \lambda$.

(B.2) AFD lift. The multi-cell register $\otimes_v A_{ZS,v}$ is an AF algebra (increasing union of finite-dimensional matrix algebras, trace-preserving inclusions). Its weak closure in the GNS representation of an emergent state ω is a hyperfinite factor: II_1 (R, unique trace) if ω is tracial, or Type III decomposing by Takesaki duality as a Type II_∞ modular crossed product if ω is non-tracial. The limit trace is the inductive-limit image of the finite canonical traces, whose normalization is fixed; the free scalar λ of (B.1) is pinned to the finite-register value, with unit $\ln(\dim Z) = \ln 2$.

(B.3) The embedding functor (F23.4). By Murray–von Neumann [13], R contains a trace-preserving copy of every finite-dimensional algebra. The functor $\Phi: A_{ZS} \rightarrow R$ restricts on the center $Z(A_{ZS}) \cong \mathbb{C}^3$ to the assignment $P_X, P_Z, P_Y \rightarrow e_X, e_Z, e_Y$ with $\tau_R(e_\alpha) = p_{eq,\alpha} = (3, 2, 6)/11$. Continuous dimension of R guarantees such projections exist; functoriality follows from the universality of R for the AF inductive system. The only residual is the identification of $(R, e_\alpha, \sigma_t^\wedge \Omega)$ with the gravitational $(M_{\text{grav}}, \text{physical projections}, \sigma_t^\wedge \text{grav})$, i.e. O-F19.6 Step 1'.

(B.4) Coarse-graining and the closure criterion (F23.5–F23.6). The conditional expectation onto the sector center is $\mathcal{E}(x) = \sum_\alpha [\tau_R(e_\alpha x)/\tau_R(e_\alpha)] e_\alpha$. It is unital, idempotent, trace-preserving, and a $Z(A_{ZS})$ -bimodule map; existence as a normal expectation follows from Takesaki [29] (automatic for the tracial state, whose modular flow is trivial). Its pushforward of the trace is $\mathcal{E}^*(\tau_R) = (3, 2, 6)/11$, giving coarse-grained $\Delta K(X \rightarrow Y) = \ln(3/6) = -\ln 2$ (F23.5). A density with continuous spectrum maps under $\mathcal{E}$ to a discrete three-outcome distribution, which is why the discrete K_Ω is consistent with a continuous gravitational modular spectrum. The closure criterion (F23.6) is the predicate $C(\omega) := [\mathcal{E}^*(\omega) = (3, 2, 6)/11]$; it is non-vacuous (true for the tracial / de Sitter maximum-entropy state, false for a generic state), so Step 1' reduces to the single falsifiable statement $C(\omega_{\text{grav}})$.

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- [31] K. Kang, ZS-M17 v1.0: Continuum-Limit Rigor for Z-Spin Lattice QFT — Reflection Positivity, OS Reconstruction (Z-Spin Cosmology, 2026). Theorem M17.1.
- [32] K. Kang, ZS-M30 v1.0: Z-Spin Duality and the RH Bridge (Z-Spin Cosmology, 2026). Theorem 30.1 (Möbius-Trace Infinity, six-route).
- [33] K. Kang, ZS-F18 v2.1: The Twelve Encounters — the Finite/Infinite Möbius-Interface Polarity (Z-Spin Cosmology, 2026). §5, §7.4.

Version History

v1.3 (June 2026): Resolves the finite–infinite edge-mode tension (v1.2 NC-F23.9) in the characteristic Z-Spin manner. Adds Theorem F23.7 (Edge-Mode Frame-Duality Interface, DERIVED-interpretation): following the ZS-F18 method (mediate finite and infinite through the $\dim(Z) = 2$ seam, not directly), the infinite KKS edge-mode tower is the external-frame count (Infinity_A) of a structure whose internal-frame closure (Infinity_B) is the $\dim(Z) = 2$ bit $\ln 2$, connected by the frame transformation of ZS-M30 Theorem 30.1, with controlling theorem the X–Y Tiling Asymmetry (ZS-M6 §5.5, PROVEN; ZS-M17, DERIVED) — the seventh route of the six-route Möbius-trace pattern (§6.1). Upgrades NC-F23.9 (OPEN tension) to DERIVED-interpretation; the residual sharpens to Condition C-edge (NC-F23.10, a patterned sub-question of Condition C). Adds gate F-F23.9, legend row DERIVED-interpretation, references [30]–[33] (ZS-M6, ZS-M17, ZS-M30, ZS-F18), and Category H (4 checks). Verification 37/37 → 41/41 PASS. F23.1, F23.2 remain DERIVED-CONDITIONAL on Condition C; F23.4–F23.7 unconditional in their registered status. No new free parameter; (A, Q, $\dim Z$) = (35/437, 11, 2) LOCKED.

v1.2 (June 2026): Deep-exploration paper on whether O-F19.6 Step 1' can be closed. Resolves the discrete-versus-continuous spectral objection by identifying K_Ω as the sector coarse-graining of the gravitational modular operator (§4.7). Adds Theorem F23.5 (Coarse-Grained Trace Matching, DERIVED, via Takesaki's conditional-expectation theorem [29]): the conditional expectation onto the sector center pushes the trace to (3, 2, 6)/11 with $\Delta K = -\ln 2$ (§4.6). Adds Lemma F23.6 (Closure Criterion, DERIVED): Step 1' closes iff Condition C — the de Sitter maximum-entropy state coarse-grains to sector weights (3, 2, 6)/11 (§4.7). Verdict: Condition C is the Z-Spin emergence dictionary and cannot be established with current tools; O-F19.6 Step 1' remains a genuine OPEN, now sharply characterized (one falsifiable statement, gate F-F23.8). Adds the finite-versus-infinite edge-mode tension to Table 2 (NC-F23.9). Verification 32/32 → 37/37 PASS (new Category G). F23.1, F23.2 remain DERIVED-CONDITIONAL on Condition C; F23.4–F23.6 unconditional. No new free parameter; (A, Q, $\dim Z$) = (35/437, 11, 2) LOCKED.

v1.1 (June 2026): Adds Theorem F23.4 (Trace-Preserving Central Embedding Functor, DERIVED) via the Murray–von Neumann universal-embedding theorem (§3.4), closing the algebraic half of O-F19.6 Step 1 unconditionally; upgrades the embedding, the central-projection traces (3,2,6)/11, and the value $\frac{1}{2} \ln 2$ to DERIVED. Sharpens the residual to O-F19.6 Step 1' (modular-flow matching) with observer-clock and edge-mode promotion paths (§4.5). Adds §6, a slot-by-slot comparison with the Klinger–Kudler–Flam–Satishchandran / KLS / CLPW edge-mode program (Table 2). Executes the pre-registered anti-numerology Monte Carlo AN-F23.1 as a joint structural-signature test: $p \approx 0.058\% \ll 1\%$ (PASS), removing the v1.0 pending status, with the inheritance caveat documented (§7.2). Verification 28/28 → 32/32 PASS. F-F23.3 now PASSING. No new free parameter; (A, Q, $\dim Z$) = (35/437, 11, 2) LOCKED.

v1.0 (June 2026): Initial public release. Established Theorem F23.1 (Finite-Register Trace Determinacy, DERIVED-CONDITIONAL), Theorem F23.2 (Z_2 -Seam Additive Constant $c = \frac{1}{2} \ln 2$), and Reading F23.3 (HYPOTHESIS). Closed Steps 2–3 of O-F19.6 at DERIVED-CONDITIONAL. Registered F-F23.1–F-F23.7 and pre-registered AN-F23.1 (then PENDING). 28/28 verification checks PASS.